

Dawei (David) Liu

Research focus: Reliable tool-using AI systems: skill retrieval, execution assurance, and multimodal workflows.

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EDUCATION

University of Pennsylvania

M.S.E. in Computer & Information Science | **GPA: 3.90/4.00**

Aug 2024 - May 2026

Philadelphia, PA

Northeastern University

B.E. in Software Engineering | **Rank: 5/396** (record: Sep 11, 2023)

Sep 2020 - Jun 2024

Shenyang, China

RESEARCH & OPEN-SOURCE EXPERIENCE

Graph-of-Skills (GoS)

EMNLP 2026 (Main Conference)

Contribution: First author and designated Core Contribution author; authored the initial public implementation and led the reported experiments. Built dependency-aware retrieval using hybrid semantic-lexical seeding, reverse-weighted Personalized PageRank, and budget-aware loading. **Results:** **25.6% higher reward** (+7.0 absolute points), **56.7% fewer total tokens** vs. full skill loading on 1,000-skill SkillsBench (GPT-5.2 Codex).

Dr. Claw

EMNLP 2026 (System Demonstrations)

System: AI scientist workspace. **Contribution:** One of the core contributors and active maintainers; 57 non-merge commits to main (#4 contributor). Authored the merged cross-platform desktop/release PR; implemented provider integrations, project/session workflows, and reliability fixes.

COMFYCLAW

2nd MUCG Workshop at ECCV 2026

System: Typed-graph harness for validation, rollback, repair, and skill evolution. **Contribution:** Equal-contribution coauthor.

PROFESSIONAL EXPERIENCE

ByteDance Inc.

Jun 2026 - Present

Software Engineer, Commerce Ads

Seattle, WA

- **Agent operations:** Built a production on-call agent that triages Lark-thread alerts using SOPs, runtime telemetry, change history, and service-topology knowledge.
- **Inference infrastructure:** Built a multi-protocol interoperability layer connecting an internal model platform to Claude Code-, Codex-, and Responses/Messages-compatible coding agents while preserving native sandboxing, approval policies, and multi-agent workflows.
- **Validation autonomy:** Built a resumable, evidence-gated validation framework with contract-defined gates, artifact/runtime checks, side-effect deduplication, and auditable ledgers.

ByteDance Inc.

May 2025 - Dec 2025

Software Engineer Intern, Commerce Ads

Seattle, WA

- Developed Shop Ads image-selection ranking with posterior features, exploration-exploitation, and multimodal LLM quality evaluation.
- Automated creative generation for high-value products through scheduled workflows and Flink-backed asset persistence; built modular preview/diagnostics with test-entity injection for ad-format and rollout validation.
- Improved serving reliability and tail latency through caching, asynchronous batch fetches, and offline feature computation.

Amazon

Jun 2024 - Aug 2024

Software Development Engineer Intern, Global-Mile Team

Beijing, China

- Built a ByteBuddy-based OpenTelemetry Java agent to instrument distributed requests without application changes, capturing payloads and traces.
- Correlated traces by business ID across partially instrumented services; dynamically selected Kinesis or SQS transport for Fargate and Lambda runtimes.
- Built a React/Java diagnostics console with searchable traces, timelines, and payload views.

JD Technology

Jul 2023 - Oct 2023

Software Engineer Intern, Algorithm Tools

Beijing, China

- Redesigned Kubernetes resource management with ConfigMaps, Informers, asynchronous updates, and sharded locks, reducing cold-start overhead.
- Modularized campaign-page generation with the Strategy pattern and generative AI; improved multi-tenant delivery with Argo Workflows, Argo CD, and Helm.

SELECTED PUBLICATIONS

- Graph-of-Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills** 2026
Dawei Liu[†], Zongxia Li[†], Hongyang Du, Xiyang Wu, Shihang Gui, Yongbei Kuang, Lichao Sun.
EMNLP 2026 (Main Conference); AgentSkills 2026 Poster, ACM CAIS. arXiv:2604.05333. [paper] [code]
Highest average reward among evaluated methods in all six SkillsBench/ALFWorld model-benchmark blocks.
- COMFYCLAW: Self-Evolving Skill Harnesses for Image Generation Workflows** 2026
Zongxia Li[†], Dawei Liu^{*}, Fuxiao Liu, Yuhang Zhou, Xiyang Wu, et al.
2nd MUCG Workshop at ECCV 2026. arXiv:2607.01709. [paper] [code]
- Dr. Claw: An AI Scientist Workspace for Vibe Research** 2026
Dingjie Song, Hanrong Zhang, Dawei Liu, Yixin Liu, Zongxia Li, Zhengqing Yuan, et al.
EMNLP 2026 (System Demonstrations). arXiv:2609.00365. [paper] [code]
- A Cookbook of 3D Vision: Data, Learning Paradigms, and Application** 2026
Hongyang Du^{*}, Zongxia Li[†], Dawei Liu^{*}, Runhao Li^{*}, Haoyuan Song, Qingyu Zhang, et al.
CVPR 2026 Workshop OpenSUN3D. pp. 8376-8390. [paper] [code]
- Multimodal Video Generation Models with Audio: Present and Future** 2026
Dawei Liu, Hongyang Du, Zongxia Li, Xiyang Wu, Lantao Yu, Jingxi Chen, et al.
ACM Multimedia 2026 Workshop JAV-CG. [paper] [code]
- TIMEDB: tumor immune micro-environment cell composition database with automatic analysis and interactive visualization** 2023
Xueying Wang^{*}, Lingxi Chen^{*}, Wei Liu^{*}, Yuanzheng Zhang, Dawei Liu, et al.
Nucleic Acids Research. 51(D1):D1417-D1424. [paper] [site]
Contribution: Developed the visualization framework and several visualization analyses. [team contribution]

^{*} Equal contribution; [†] Core Contribution, as designated by the cited papers.

RESEARCH IMPACT & SERVICE

- Citations.** 88 citations across six selected works (Sep 14, 2026).
- Research use.** GoS: 14 source-checked papers document public-asset use, implementation or reimplementa-tion, controlled comparison, or component reuse and testing (Sep 12, 2026).
- Open-source interest.** GoS: 208 stars, 26 forks; Dr. Claw: 1,105 stars, 123 forks (Sep 14, 2026).
- Peer review.** ACM CAIS Workshop AgentSkills (2026): Completed reviews of two assigned papers; service acknowledged by the organizing committee.

SELECTED HONORS

- Winner, Apple Swift Student Challenge** 2026
Built **Aura Chef**, a touchless Swift playground integrating hand-pose recognition and voice commands; selected as one of 350 winners from 37 countries and regions.
- China Collegiate Computing Contest - Mobile Application Innovation** 2022
Team lead & sole software developer of *Trace Note* (iOS/AR); National Final Second Prize and Northeast Division Innovation Track First Prize.
- National Scholarship for Undergraduate Students** 2020-2021
Ministry of Education, China.

TECHNICAL SKILLS

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| Languages | Python, Go, Java, C/C++, JavaScript/TypeScript, SQL, Swift |
| Agent/ML | LLM agents, tool calling, retrieval/GraphRAG, multimodal evaluation, PyTorch, ranking/recommendation |
| Systems/Infra | Distributed services, OpenTelemetry, Kafka, Flink, MySQL, Redis, Elasticsearch, Thrift/Protobuf |
| Cloud/DevOps | Linux, Docker, Kubernetes, AWS, Helm, CI/CD, Grafana, Git |